GlucoTwin Bharat · System architecture

Type 2 diabetes digital twin: two data streams → team of experts → LLM-routed fusion → doctor dashboard. No model trained from scratch.

1 · DATA STREAMS

Static: health record

- Demographics, waist, BMI
- ICD-10 diagnoses
- Labs: HbA1c, fasting glucose & insulin, lipids, eGFR
- Medicines, BP
- Family history, TCF7L2 genotype, PRS

Synthetic FHIR-style JSON

Dynamic: wearables

- CGM every 5 min
- Heart rate, steps
- Sleep stages & hours
- Raw PPG pulse waveform
- Meal log (Indian meal library with GI)

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2 · EXPERTS (zero-shot, all on device)

Physiology twin + data assimilation

Bergman minimal model + gut absorption, insulin secretion, exercise, liver suppression, dawn effect. Parameters from EHR via HOMA-IR / HOMA-B, metformin & sulfonylurea effects, TCF7L2. Sleep & HRV modulate insulin sensitivity. Every 2 h re-fits insulin sensitivity θ and basal shift ΔGb to CGM → **twin divergence**. Simulates **what-if** plans.

Chronos-Bolt foundation model

Amazon's pretrained time-series model, zero-shot 2-h glucose forecast with quantiles.

PPG expert: PaPaGei-S + beat detection

HR & HRV computed from raw pulse; 512-d PaPaGei embedding → nightly waveform-shape drift.

Autonomic expert

Personal-baseline robust z-scores of HRV, resting HR, sleep → cardiovascular strain flag.

Clinical risk equations

Framingham 10-yr CVD · Indian Diabetes Risk Score · CGM consensus (TIR, GMI, CV, LBGI/HBGI).

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3 · FUSION & LEARNING

LLM router + skill fusion

Each expert graded on 16 hindcasts over the last 8 h ($1/MAE^2$). LLM router reads context (meal just logged, night, drift) and explains trust; sanity guard overrides bad routing. Weights = ½ skill + ½ router.

Calibrated alerts

Split-conformal 80% band → P(>180), P(<70) in 2 h. Alerts: spike, high, hypo, twin divergence, autonomic strain.

Online adapter `on device`

192 weights, recursive least squares, learns the patient's own error; self-gating.

Federated learning

FedAvg over adapter weights only, update clipping + Gaussian noise (differential privacy) → global prior for new patients.

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4 · DOCTOR & LLM LAYER

Privacy gateway

Names/IDs/city → pseudonyms; Aadhaar, ABHA, PAN, phone, email scrubbed; age → band; every payload audit-logged. DPDP Act by design.

LLM (DeepInfra Llama-3.1-8B or local Ollama)

Router · grounded patient chat · clinical brief · what-if parser.

FastAPI + React dashboard

Live replay clock, auto-refresh every 2 s. Tabs: chat, ask the twin, twin vs record, sleep & heart, record, brief, learning, privacy.

EDGE All models, raw data and the adapter stay on the device. Only de-identified text can reach a cloud LLM — or none with a local LLM. Only 192 clipped + noised weights reach the federated server.

GlucoTwin Bharat · Data flow, tech stack & deployment

Request flow (every 5 simulated minutes)

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Wearable sample arrives (CGM, HR, steps, sleep, PPG, meal)
   │
   ├─► Physiology twin: assimilate last 8 h → θ, ΔGb → 2-h forecast
   ├─► Chronos-Bolt: zero-shot 2-h forecast (q10/q50/q90)
   ├─► PPG / autonomic: nightly HRV, RHR, sleep, pulse-shape drift
   │
   ├─► Hindcasts (16 origins, last 8 h) → MAE per expert → skill weights
   ├─► LLM router (via privacy gateway) → context weights + reason
   │
   ├─► Fused forecast + online adapter correction
   ├─► Split-conformal band → P(>180), P(<70)
   └─► Alerts → /api/state → React dashboard (poll 2 s)

Doctor actions
   what-if  → parse (LLM) → re-simulate 4 h (plan A vs B)
   chat     → live-twin context + optional simulation → LLM
   brief    → twin numbers → LLM → SUMMARY / ALERTS / ACTIONS
```

Technical stack

Layer	Technology
Language / numerics	Python 3.10+, NumPy, SciPy, pandas
Backend	FastAPI, Uvicorn, Pydantic
Glucose foundation model	Amazon Chronos-Bolt (PyTorch, Hugging Face)
PPG foundation model	PaPaGei-S, Nokia Bell Labs (PyTorch)
LLM	DeepInfra Llama-3.1-8B-Instruct (OpenAI-compatible) / local Ollama
Frontend	React 18, Vite 5, Recharts 2
Privacy	Regex + salted pseudonymisation gateway, egress audit log
Learning	Recursive least squares adapter, FedAvg + DP noise
Data	Synthetic EHR + wearable generator (DPDP / HIPAA sandbox)

API

GET /api/state	everything the dashboard shows
POST /api/whatif	4-h counterfactual simulation
POST /api/chat · /api/brief	grounded LLM chat and clinical brief
GET /api/privacy · /api/learning	egress log · adapter & federated stats
POST /api/clock	live replay control